

4 Lipids and carbohydrates can be used as energy storage molecules.

(a) Lactose is an important energy source in milk.

(i) What type of chemical reaction is involved in breaking a lactose molecule into two monosaccharides?

(1)

- ☐ **A** condensation
- ☐ **B** esterification
- ☐ **C** hydrogen bonding
- ☐ **D** hydrolysis

(ii) Milk contains 4.9 g of lactose per 100 g of milk.

Each gram of lactose provides 16 kJ of energy.

Calculate the energy available from the lactose in 200 g of milk.

(1)

Answer kJ

(b) Carbohydrates used as energy storage molecules include glycogen and starch.

Compare and contrast the structures of glycogen and starch.

(4)

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- (c) (i) State one difference between the structure of a saturated lipid and an unsaturated lipid.

(1)

- (ii) Describe how a triglyceride is synthesised.

(3)

(Total for Question 4 = 10 marks)